BCSE429P

Cyber Physical Systems Design Lab

Laboratory 1 - MQTT Using Node JS with HiveMQ Serverless Cloud

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Question:

Design an MQTT Publisher Subscriber module using Node JS as per the following scenario:

There are three publishers publishes topics, namely "Milk", "Coffee" and "Water". Based on the percentage of coffee, milk and water, the subscriber will get a cappuccino, espresso, or mochaccino.

Design a logic to build the above scenario. This application suits a robot that can prepare cappuccino, espresso, or mochaccino based on the quantity of milk, coffee and water.

subscriber.js

```
const mqtt = require("mqtt");

const options = {
    host: "f18218ea265145ff994b464c4a8d86cb.s1.eu.hivemq.cloud",
    port: 8883,
    protocol: "mqtts",
    username: "amansingh",
    password: "Aman@8409"
};

const client = mqtt.connect(options);

let milk = 0;
let coffee = 0;
let water = 0;

client.on("connect", () => {

    console.log("Subscriber Connected");

    client.subscribe("Milk");
    client.subscribe("Coffee");
    client.subscribe("Water");

});

client.on("message", (topic, message) => {

    const value = parseInt(message.toString());

    if (topic === "Milk")
        milk = value;

    if (topic === "Coffee")
        coffee = value;

    if (topic === "Water")
        water = value;

    console.clear();

    console.log("Milk   :", milk + "%");
    console.log("Coffee :", coffee + "%");
    console.log("Water  :", water + "%");
```

```
    console.log("-------------------------");

    // Decision Logic

    if (coffee >= 60 && milk <= 30) {
        console.log("☕ Espresso Prepared");
    }
    else if (milk >= 60 && coffee >= 30 && water <= 30) {
        console.log("☕ Cappuccino Prepared");
    }
    else if (milk >= 40 && coffee >= 40 && water >= 20) {
        console.log("☕ Mochaccino Prepared");
    }
    else {
        console.log("Waiting for suitable ingredient percentages...");
    }

});
```

milkPublisher.js

```
const mqtt = require("mqtt");

const options = {
    host: "f18218ea265145ff994b464c4a8d86cb.s1.eu.hivemq.cloud",
    port: 8883,
    protocol: "mqtts",
    username: "amansingh",
    password: "Aman@8409"
};

const client = mqtt.connect(options);

client.on("connect", () => {
    console.log("Milk Publisher Connected");

    setInterval(() => {

        let milk = Math.floor(Math.random() * 51) + 30; //30-80%

        console.log("Publishing Milk :", milk);

        client.publish("Milk", milk.toString());
```

```
    }, 5000);
});
```

coffeePublisher.js

```
const mqtt = require("mqtt");

const options = {
    host: "f18218ea265145ff994b464c4a8d86cb.s1.eu.hivemq.cloud",
    port: 8883,
    protocol: "mqtts",
    username: "amansingh",
    password: "Aman@8409"
};

const client = mqtt.connect(options);

client.on("connect", () => {

    console.log("Coffee Publisher Connected");

    setInterval(() => {

        let coffee = Math.floor(Math.random() * 81) + 20;

        console.log("Publishing Coffee :", coffee);

        client.publish("Coffee", coffee.toString());

    }, 5000);

});
```

waterpublisher.js

```
const mqtt = require("mqtt");

const options = {
    host: "f18218ea265145ff994b464c4a8d86cb.s1.eu.hivemq.cloud",
    port: 8883,
    protocol: "mqtts",
    username: "amansingh",
    password: "Aman@8409"
};
```

```
const client = mqtt.connect(options);

client.on("connect", () => {

    console.log("Water Publisher Connected");

    setInterval(() => {

        let water = Math.floor(Math.random() * 41) + 20;

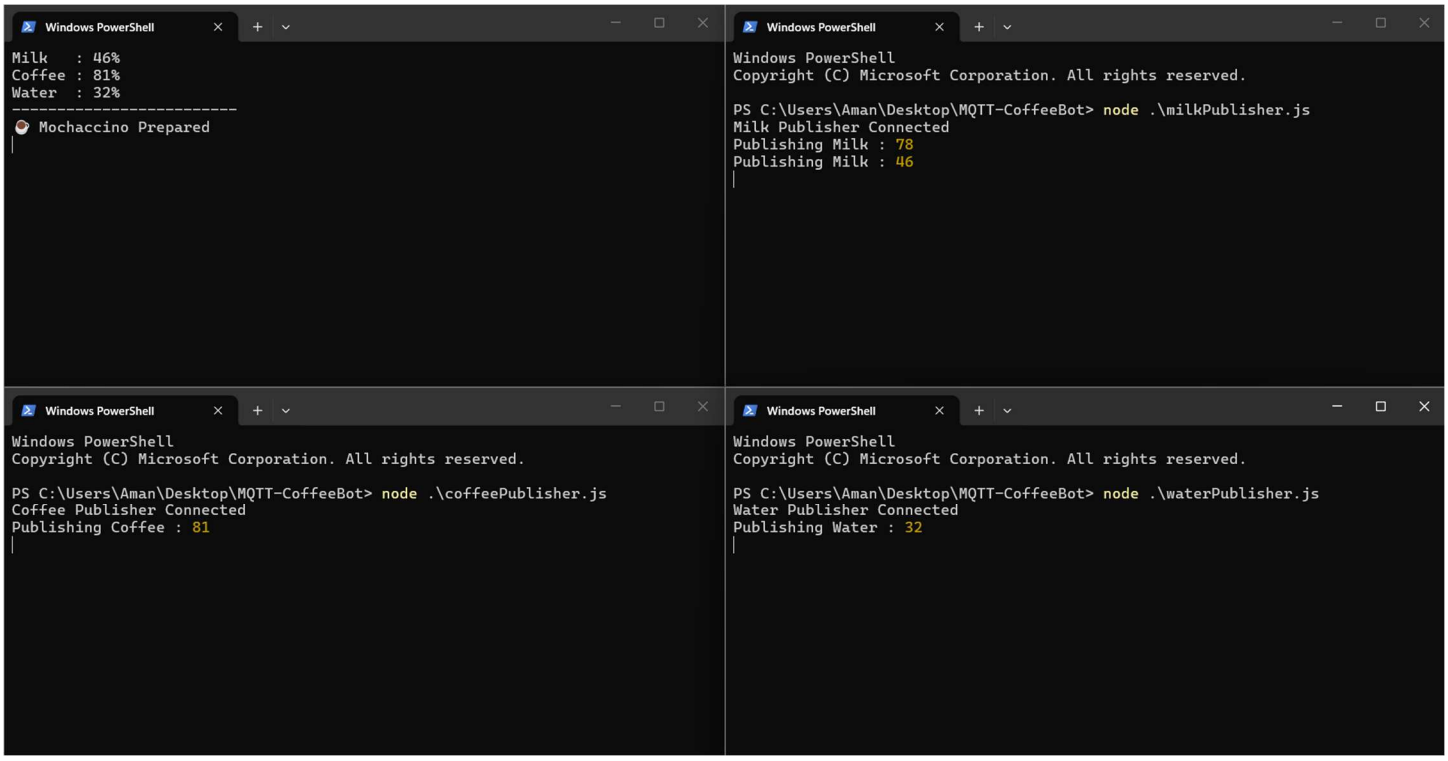
        console.log("Publishing Water :", water);

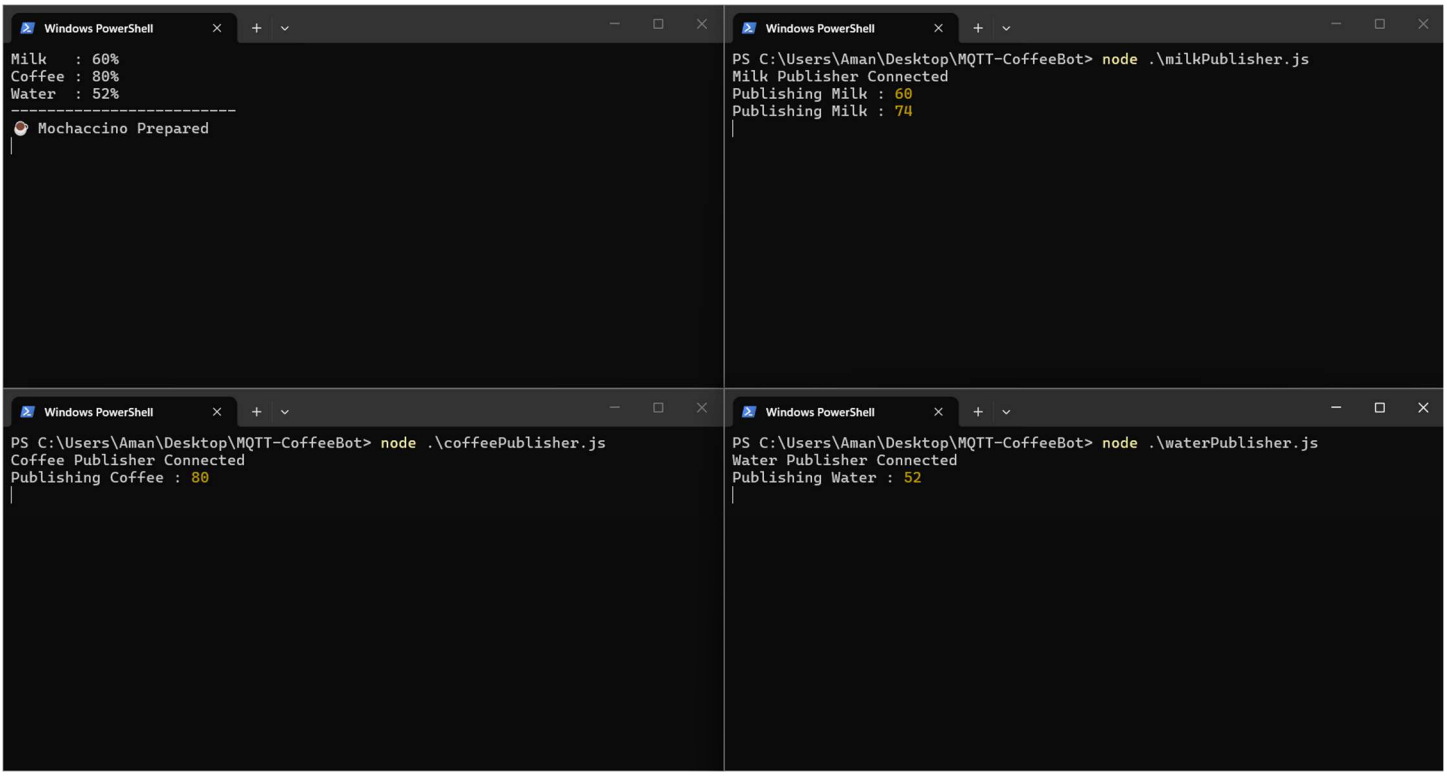
        client.publish("Water", water.toString());

    }, 5000);

});
```

Output:




Windows PowerShell
Milk : 60%
Coffee : 80%
Water : 52%
Mochaccino Prepared
Windows PowerShell
PS C:\Users\Aman\Desktop\MQTT-CoffeeBot> node .\milkPublisher.js
Milk Publisher Connected
Publishing Milk : 60
Publishing Milk : 74
Windows PowerShell
PS C:\Users\Aman\Desktop\MQTT-CoffeeBot> node .\coffeePublisher.js
Coffee Publisher Connected
Publishing Coffee : 80
Windows PowerShell
PS C:\Users\Aman\Desktop\MQTT-CoffeeBot> node .\waterPublisher.js
Water Publisher Connected
Publishing Water : 52